

SSH KEY RECOVERY & EXPLOITATION

HEX-MAGIC & CRYPTANALYSIS FIELD GUIDE // SCARLET OPERATIONS

SECTION 13 — CRACKING AN ENCRYPTED SSH PRIVATE KEY

If an SSH private key is protected with a passphrase, John the Ripper cannot directly use the original key file. First, convert the SSH private key into a hash format that John understands using `ssh2john`.

IMPORTANT AUTHORIZATION NOTICE: Only perform password/passphrase recovery on SSH keys that you own or are explicitly authorized to test, such as in a CTF or training lab environment.

13.1 STEP 1 — EXTRACT THE HASH

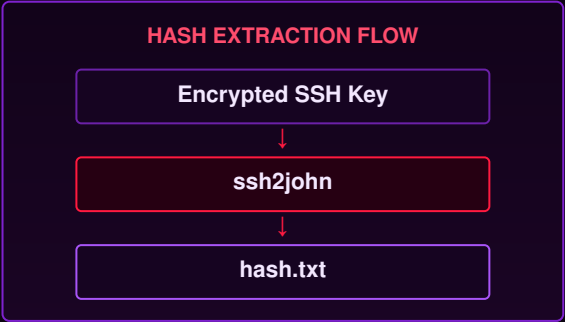
Convert the encrypted SSH private key into a readable hash file:

```
ssh2john id_rsa_1593558668558.id_rsa > hash.txt
```

This creates `hash.txt` containing the extracted data in a format John the Ripper can process.

If `ssh2john` Is Not in PATH (Kali Linux):

```
python3 /usr/share/john/ssh2john.py  
id_rsa_1593558668558.id_rsa > hash.txt
```



13.2 STEP 2 — CRACK THE PASSPHRASE

Run John the Ripper against the extracted hash using a wordlist:

```
john --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
```

COMMAND COMPONENT	DESCRIPTION / PURPOSE
<code>john</code>	Executes the John the Ripper cracking engine
<code>--wordlist=</code>	Specifies the wordlist mode and target dictionary file
<code>/usr/share/wordlists/rockyou.txt</code>	Standard Kali Linux common password dictionary
<code>hash.txt</code>	Target file containing the extracted SSH key hash

13.3 STEP 3 — DISPLAY THE CRACKED PASSPHRASE

After John completes the cracking process, view the recovered credential:

```
john --show hash.txt
```

This displays the recovered passphrase if John successfully discovered a match in the wordlist.



SECTION 14 — USE THE RECOVERED SSH KEY

Once you possess the passphrase, use the private key for SSH authentication to the target system.

STEP 1 — SET STRICT PERMISSIONS

SSH clients require private keys to have restricted owner-only access permissions:

STEP 2 — CONNECT USING PRIVATE KEY

Authenticate to the target host specifying the key file:

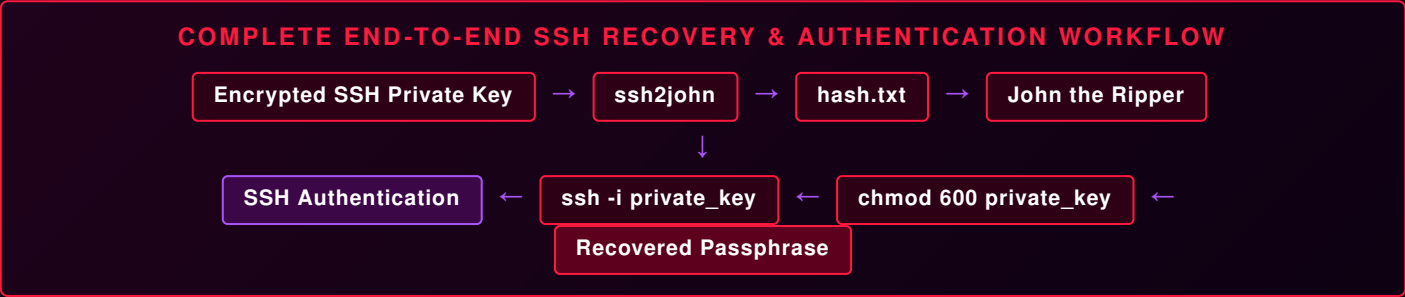
```
chmod 600 id_rsa_1593558668558.id_rsa
```

PERMISSIONS SYNTAX	<code>-rw-----</code>
OWNER ACCESS	Read + Write
GROUP / OTHERS	No Permissions

```
ssh -i id_rsa_1593558668558.id_rsa user@target_ip
```

When prompted, enter the recovered passphrase:

Enter passphrase for key 'id_rsa...':
[RECOVERED_PASSPHRASE]



QUICK COMMAND CHEAT SHEET

1. EXTRACT HASH

```
ssh2john id_rsa_1593558668558.id_rsa > hash.txt
```

2. RUN JOHN

```
john --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
```

3. SHOW RESULT

```
john --show hash.txt
```

4. FIX PERMISSIONS

```
chmod 600 id_rsa_1593558668558.id_rsa
```

5. SSH LOGIN

```
ssh -i id_rsa_1593558668558.id_rsa username@target_ip
```

IMPORTANT CONCEPTUAL DIFFERENCE

1. Private-Key Passphrase

Protects the encrypted private key stored on local disk.
Private Key → Encrypted/Protected → Passphrase Required

2. SSH Login Authentication

Once unlocked, the key pair authenticates against the remote server's authorized keys.
Private Key + Authorized Public Key → SSH Authentication

Summary Workflow:

Crack passphrase → Unlock key → Authenticate to SSH server.